

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a2808d9ba617642b5.jsonl`  
- SHA-256: `cfb7fca399794a013c95ac88001472b876be4b81053f5a86b6c6db3785d72b64`  
- Source modified: `2026-06-16T03:26:03+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:24:37.194Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research question: ****What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?**** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best ***recent*** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"In the ablation, a learned 1-D convolution TCM block is the WORST option (59.4% avg mAP, 7.4 points below ActionFormer's 66.8), which the authors attribute to its having the most parameters and overfitting on informative-but-redundant clip features; parameter-free Subsampling (61.0) and Average Pooling (63.2) both beat convolution, and MaxPooling (67.7) is best ? directly evidence that for small/redundant TAL data, fewer/zero learned temporal parameters generalize better."

****Source:**** <https://arxiv.org/abs/2303.09055> (primary)

****Supporting quote:**** "Conv[28](OurImpl) ... 59.4 ... the convolution layer introduces the most parameters which tend to overparameterize the model. We argue that given the informative features extracted from the pretrained 3D CNN, the model should not contain too many parameters, which may lead to overfitting and thus less generalization."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:24:40.088Z)

I'll verify this claim about the TCM ablation study in arXiv 2303.09055. Let me start by identifying the paper and checking the specific numbers.

3. assistant (2026-06-16T03:24:46.066Z)

I'll fetch the paper's abstract first to confirm what it is, then dig into the ablation table.

4. user (2026-06-16T03:24:51.505Z)

Paper Summary

Title: "TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization"

Authors: Tuan N. Tang, Kwonyoung Kim, Kwanghoon Sohn

Venue/Year: arXiv (Computer Vision and Pattern Recognition), submitted March 16, 2023

Main Topic: Yes, this work directly addresses Temporal Action Localization (TAL) and temporal context modeling.

Abstract Summary

The researchers propose TemporalMaxer, a streamlined approach to TAL that challenges the prevailing trend of using complex architectures. Rather than employing sophisticated temporal context modeling (TCM) mechanisms like self-attention, they argue that video features already contain sufficient information. Their method employs a "basic, parameter-free, and local region operating max-pooling block" to extract critical information from adjacent clip embeddings.

Key Claims: The approach outperforms state-of-the-art methods utilizing long-term TCM while requiring significantly fewer parameters and computational resources. The authors provide publicly available code for their implementation.

5. user (2026-06-16T03:24:55.091Z)

Web search results for query: "arXiv 2303.09055 TCM temporal action localization ablation MaxPooling Conv ActionFormer"

Links: [{"title": "arXiv:2303.09055v1 [cs.CV] 16 Mar 2023", "url": "https://arxiv.org/pdf/2303.09055"}, {"title": "GitHub - TuanTNG/TemporalMaxer: TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization", "url": "https://github.com/TuanTNG/TemporalMaxer"}, {"title": "GitHub - f1brahim-tmu/TemporalMaxer: TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization", "url": "https://github.com/f1brahim-tmu/TemporalMaxer"}, {"title": "[2303.09055] TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization", "url": "https://arxiv.org/abs/2303.09055"}, {"title": "ContextDet: Temporal Action Detection with Adaptive Context Aggregation", "url": "https://arxiv.org/html/2410.15279v1"}, {"title": "Temporal Action Localization with Cross Layer Task Decoupling and Refinement", "url": "https://arxiv.org/html/2412.09202v1"}, {"title": "Temporal Action Detection Model Compression by Progressive Block Drop", "url": "https://arxiv.org/html/2503.16916v1"}, {"title": "End-to-End Temporal Action Detection with 1B Parameters Across 1000 Frames", "url": "https://arxiv.org/html/2311.17241v2"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/pdf/2512.01298"}]

Based on the search results, I found information about arXiv paper 2303.09055. Here's what I found:

The paper introduces TemporalMaxer, which minimizes long-term temporal context modeling while

maximizing information from extracted video clip features with a basic, parameter-free, and local region operating max-pooling block.

TemporalMaxer is effective at modeling temporal contexts, which outperforms the robust baseline, ActionFormer, with 2.8x fewer GMACs and 3x faster inference speed. When comparing only the backbone time, the proposed method only takes 2.5 ms which is 8.0x faster than ActionFormer backbone, 20.1 ms.

Regarding the ablation studies mentioned in your query, the paper presents results of other blocks other than Max Pooling in ablation studies about different TCM blocks on THUMOS14. The motivation comes from PoolFormer that replaces computationally intensive and highly parameterized attention modules with the most basic block in deep learning, the pooling layer.

The paper demonstrates that a simple max-pooling approach can be surprisingly effective for temporal action localization compared to more complex architectures like ActionFormer's Transformer-based approach.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:24:58.696Z)

The paper is confirmed: TemporalMaxer (arXiv 2303.09055). The thesis directly supports the claim's spirit (parameter-free max pooling beats heavier TCM). Now I need to verify the specific ablation numbers in the claim: Conv=59.4, Subsampling=61.0, AvgPool=63.2, MaxPool=67.7, and the ActionFormer=66.8 comparison. Let me fetch the full PDF/HTML ablation table.

7. user (2026-06-16T03:25:05.818Z)

I cannot locate the specific ablation study table you're requesting in the provided PDF content. While the document appears to contain technical content about temporal context modules and action detection, the compressed/encoded nature of the PDF stream makes it impossible for me to reliably extract the exact numerical values for the Conv, Subsampling, Average Pooling, Max Pooling, and ActionFormer mAP comparisons on THUMOS14.

To obtain these specific numbers and the accompanying explanation about convolution's performance limitations, I recommend:

1. Accessing the paper directly in readable format
2. Checking the supplementary materials section
3. Reviewing the results table in the main paper body

I cannot provide potentially inaccurate figures when precision is essential for your research needs.

[Binary content (application/pdf, 1.1MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579965172-5ov5x2.pdf]

8. assistant (2026-06-16T03:25:07.574Z)

The PDF was saved locally. Let me read the ablation table directly from the PDF.

9. user (2026-06-16T03:25:08.603Z)

PDF file read: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579965172-5ov5x2.pdf (1.1MB)

10. assistant (2026-06-16T03:25:30.549Z)

I now have the full paper including Table 5 (the ablation) and the surrounding text. Let me verify each component of the claim against the primary source.

****Table 5 (page 8) ? exact values:****

- Conv [28] (Our Impl): 0.5=62.8, 0.7=37.1, ****Avg=59.4****
- Subsampling: Avg=****61.0****
- Average Pooling: Avg=****63.2****
- Transformer [51]: Avg=****66.8**** (this is the ActionFormer baseline)
- TemporalMaxer (MaxPooling): Avg=****67.7****

****Now checking each claim component:****

1. "Conv is the WORST option (59.4% avg mAP)" ? TRUE. 59.4 is the lowest in Table 5.
2. "7.4 points below ActionFormer's 66.8" ? TRUE. $66.8 - 59.4 = 7.4$. Confirmed both by arithmetic and the paper's own text: "the convolution layer decreases by 7.4% mAP compared with the ActionFormer baseline." And ActionFormer's avg mAP is 66.8 (Table 1 and Table 5, where the Transformer row = ActionFormer baseline).
3. "authors attribute to its having the most parameters and overfitting on informative-but-redundant clip features" ? TRUE. The quote is verbatim: "the convolution layer introduces the most parameters which tend to overparameterize the model... should not contain too many parameters, which may lead to overfitting and thus less generalization." Note: the paper gives TWO reasons ? (a) fixed conv weights can't retain most-informative features, (b) overparameterization/overfitting. The claim emphasizes the overfitting reason, which is one of the two stated reasons, and accurately frames it as "informative-but-redundant" features (the paper repeatedly stresses clip features are "informative but highly similar"). Slight nuance: the claim collapses to the overfitting reason, but it's faithfully one of the authors' two arguments.
4. "parameter-free Subsampling (61.0) and Average Pooling (63.2) both beat convolution" ? TRUE. $61.0 > 59.4$ and $63.2 > 59.4$. Confirmed by Table 5 and text.
5. "MaxPooling (67.7) is best" ? TRUE. 67.7 is the highest in Table 5; text: "Max Pooling achieves the highest results at every tIoU threshold."
6. The interpretive conclusion "for small/redundant TAL data, fewer/zero learned temporal parameters generalize better" ? This is the claim-writer's synthesis. Let me check for overreach.

****Now the adversarial part.**** Let me scrutinize the interpretive leap and check #params to ensure "most parameters" is accurate.

From Table 5, #params (M): Conv=30.5, Subsampling=7.1, AvgPool=7.1, Transformer=29.3, TemporalMaxer=7.1. So Conv (30.5M) does have MORE params than Transformer (29.3M) ? "most parameters" is literally TRUE.

Let me probe the one place where the claim could overreach: the phrase "small data." THUMOS14 has 200 validation videos (used for training in this ablation, on train+val of THUMOS14). That is NOT "small data" in the sense of the research question's ~6-16 videos. Let me verify whether the claim's editorializing ("for small/redundant TAL data") is the paper's framing or the claim-writer's extrapolation.

The paper's actual argument is about feature **redundancy** (informative-but-highly-similar 3D-CNN features) leading to overparameterization risk ? NOT about dataset size being small. THUMOS14 is a standard mid-size TAL benchmark, not a tiny dataset. The claim conflates "redundant features ? overparameterization" (the paper's point) with "small dataset ? overparameterization" (an extrapolation). Let me check this distinction is real by re-reading the exact overfitting passage.

The paper says: "given the informative features extracted from the pre-trained 3D CNN, the model should not contain too many parameters, which may lead to overfitting and thus less generalization." The overfitting argument is tied to ****feature informativeness/redundancy****, not to dataset size. The claim's phrase "small/redundant TAL data" bundles a true word (redundant) with an unsupported one (small).

Let me do a quick sanity check that no later correction/erratum changes these numbers, and confirm this is the canonical version.

11. user (2026-06-16T03:25:37.677Z)

Web search results for query: "TemporalMaxer THUMOS14 ablation convolution 59.4 subsampling 61.0 average pooling 63.2 max pooling 67.7"

Links: [{"title": "TemporalMaxer: Maximize Temporal Context with only Max Pooling", "url": "https://arxiv.org/pdf/2303.09055"}, {"title": "Temporal Max Pooling for Sequence Modeling", "url": "https://www.emergentmind.com/topics/temporal-max-pooling"}, {"title": "Forcing the Whole Video as Background: An Adversarial Learning Strategy for Weakly Temporal Action Localization", "url": "https://arxiv.org/pdf/2207.06659"}, {"title": "GitHub - flibrahim-tmu/TemporalMaxer: TemporalMaxer: Maximize Temporal Context with only Max Pooling for Temporal Action Localization · GitHub", "url": "https://github.com/flibrahim-tmu/TemporalMaxer"}, {"title": "Non-Local Temporal Difference Network for Temporal Action Detection - PMC", "url": "https://pmc.ncbi.nlm.nih.gov/articles/PMC9655564/"}, {"title": "A Comprehensive Study on Temporal Modeling for Online Action Detection", "url": "https://arxiv.org/pdf/2001.07501"}, {"title": "Difference between MaxPooling and SubSampling · Issue #944 · torch.nn", "url": "https://github.com/torch/nn/issues/944"}]

Perfect! I found the exact information you were looking for.

The search results show an ablation study from the TemporalMaxer paper examining different temporal context modeling (TCM) blocks on THUMOS14, with the following results: Subsampling achieved 61.0, Average Pooling achieved 63.2, and Max Pooling achieved 67.7.

The study notes that subsampling is prone to losing crucial information as only half of the video clip embeddings are kept after a TCM block, so the researchers replaced Transformer with Average Pooling with kernel size 3 and stride 2, which improved results by 2.2%. They then employed Max Pooling as TCM, which achieved the highest results at every tIoU threshold.

The source is the official TemporalMaxer paper, which you can find at:
<https://arxiv.org/pdf/2303.09055>

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

12. assistant (2026-06-16T03:25:49.368Z)

All numbers are confirmed against the primary source (Table 5, page 8 of arXiv:2303.09055), and the search corroborates them with no contradicting evidence or erratum.

****Verdict:**** The factual core of the claim is fully and accurately supported by the primary source:

- Conv = 59.4 (worst), Subsampling = 61.0, AvgPool = 63.2, Transformer/ActionFormer = 66.8, MaxPool = 67.7 ? all verbatim from Table 5.
- "7.4 points below ActionFormer" ? confirmed verbatim in the paper's text and arithmetic.
- "most parameters and overfitting" attribution ? verbatim quote; Conv has 30.5M params (most in the table, vs Transformer's 29.3M).
- Both parameter-free options beating Conv, MaxPool best ? all true.

****The one weak spot (adversarial scrutiny):**** The interpretive tagline "for small/redundant TAL data, fewer/zero learned temporal parameters generalize better" mildly overreaches on the word *small*. The paper's overfitting argument is grounded in feature ****redundancy/informativeness**** (pre-extracted 3D-CNN features are "informative but highly similar"), NOT dataset size. THUMOS14's train+val (~200 videos) is a standard mid-size benchmark, not "small data" in the sense of the research project's ~6-16 videos. The "redundant" half of the phrasing is well-supported; the "small" half is an extrapolation the paper does not make. However, this is the claim-writer's framing/editorial conclusion, not a factual misattribution to the authors ? and the parameter-count/overfitting mechanism is plausibly transferable to small-data settings (the direction is defensible). The supporting quote itself is accurate and load-bearing.

Since every numeric and attributional fact is verbatim-correct from the primary source, and the only stretch is a soft interpretive label (not a fabricated number or misquote), this does not rise to refutation. The claim is well-supported, current (2023, still canonical), and primary-sourced.

13. user (2026-06-16T03:25:58.896Z)

Structured output provided successfully

14. assistant (2026-06-16T03:26:02.916Z)

I have completed the adversarial verification. The claim about the TemporalMaxer (arXiv:2303.09055) TCM ablation is ****NOT refuted**** ? every numeric and attributional fact checks out verbatim against Table 5 of the primary source, with the only caveat being a soft interpretive label ("small data") that is the claim-writer's transferability framing rather than a factual error.